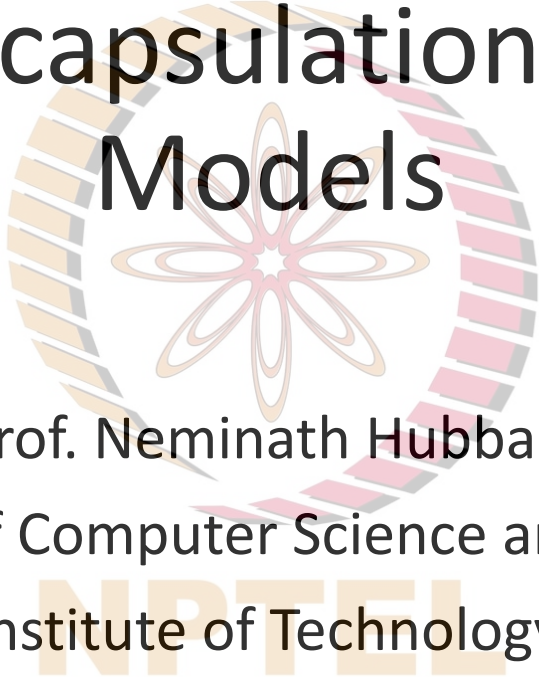


Layering, Encapsulation and Service Models



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Agenda

- ☐ Layering architecture motivating analogy
- ☐ Service models
 - ☐ Necessity
- ☐ OSI Reference Model
- ☐ TCP/IP Model
- ☐ Network Security



Protocol “layers” and Reference Models

Networks are complex,
with many “pieces”:

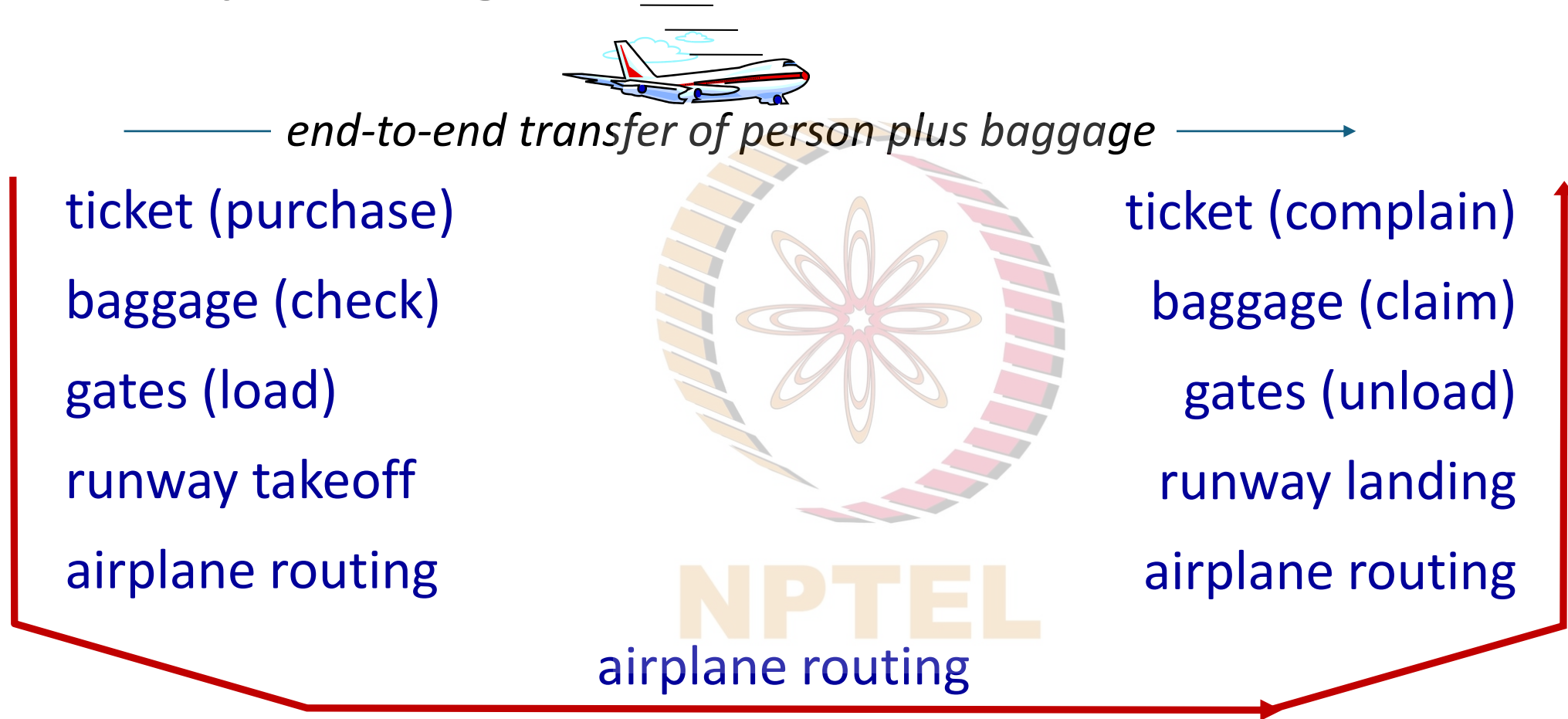
- ☐ hosts
- ☐ routers
- ☐ links of various media
- ☐ applications
- ☐ protocols
- ☐ hardware, software

Question: is there any
hope of *organizing*
structure of network?

- ☐ and/or our *discussion*
of networks?



Example: Organization of Air Travel



How would you *define/discuss* the *system* of airline travel?

- a series of steps, involving many services

Example: Organization of Air Travel



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

Why Layering?

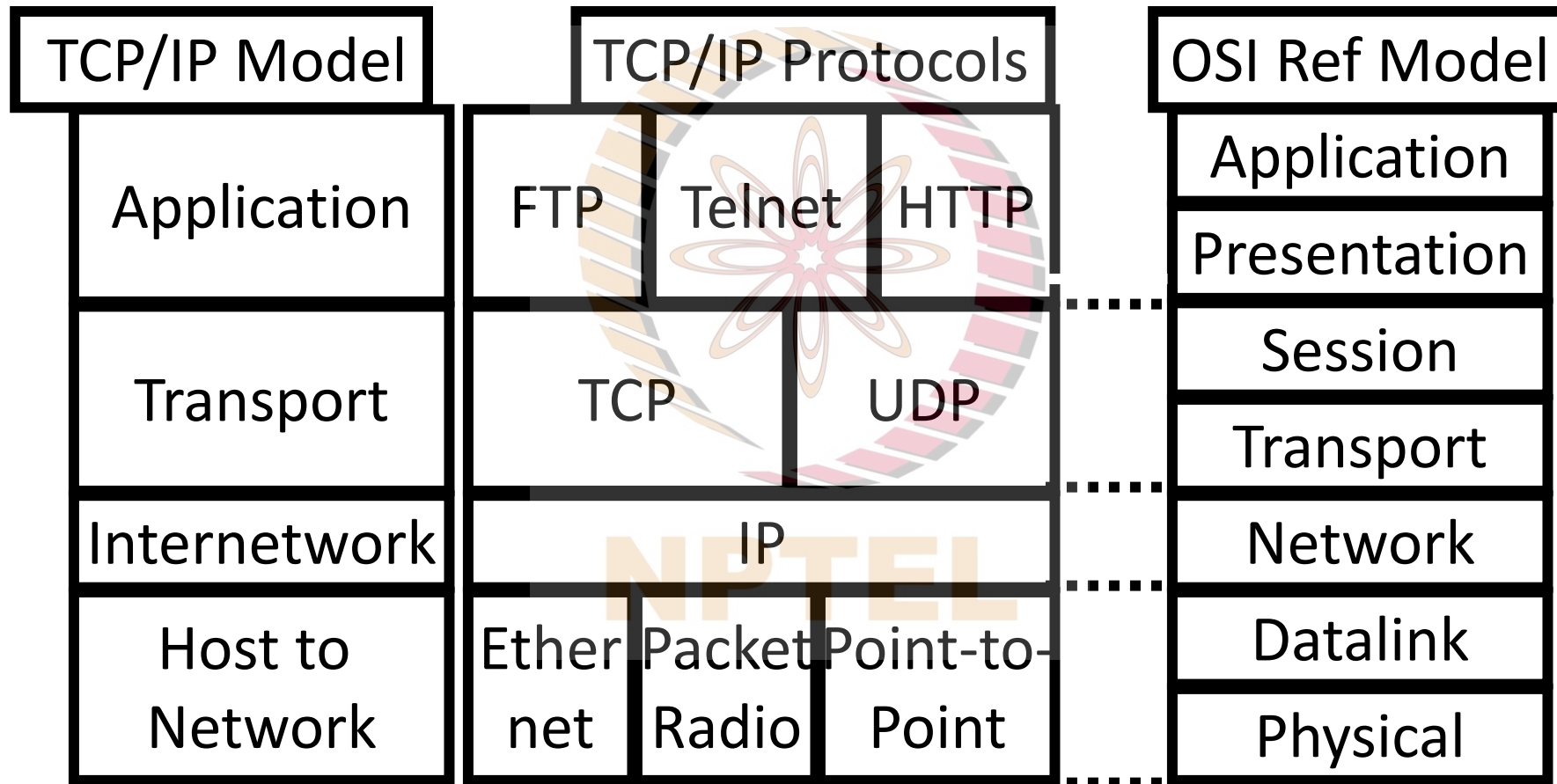
Approach to designing/discussing complex systems:

- ❑ Explicit structure allows identification, relationship of system's pieces
 - ❑ layered *reference model* for discussion
- ❑ Modularization eases maintenance, updating of system
 - ❑ change in layer's service *implementation*: transparent to rest of system
 - ❑ e.g., change in gate procedure doesn't affect rest of system

Early Networking Challenges

- ❑ Proprietary systems (no standards)
 - ❑ Interoperability became difficult
- ❑ Vendors used **different protocols**
- ❑ Networks couldn't communicate easily
- ❑ Need for a **universal framework**

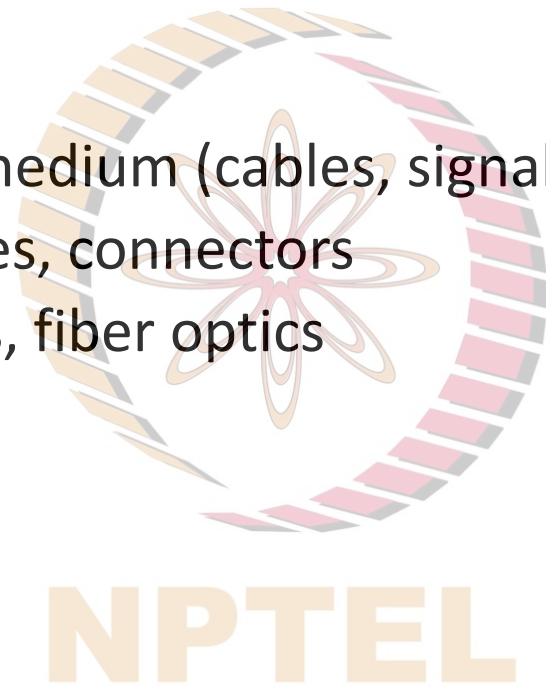
Reference Models for Layering



OSI Layers-Physical Layer

☐ Physical Layer

- ☐ Transmits raw bits over medium (cables, signals)
- ☐ Defines voltage, data rates, connectors
- ☐ Example: Ethernet cables, fiber optics



OSI Layers -Data Link Layer

☐ Data Link Layer

- ☐ Ensures **node-to-node delivery**
- ☐ Error detection & correction
- ☐ Frame formatting & MAC addressing



OSI Layers - Network Layer

☐ Network Layer

- ☐ Handles **routing and path selection**
- ☐ Logical addressing (IP addresses)
- ☐ Determines best route for data



OSI Layers - Transport Layer

☐ Transport Layer

- ☐ End-to-end communication
- ☐ Reliability (error recovery, retransmission)
- ☐ Flow control & segmentation



OSI Layers –Session Layer

☐ Session Layer

- ☐ Manages sessions (start, maintain, terminate)
- ☐ Synchronization between devices



OSI Layers –Presentation Layer

☐ Presentation Layer

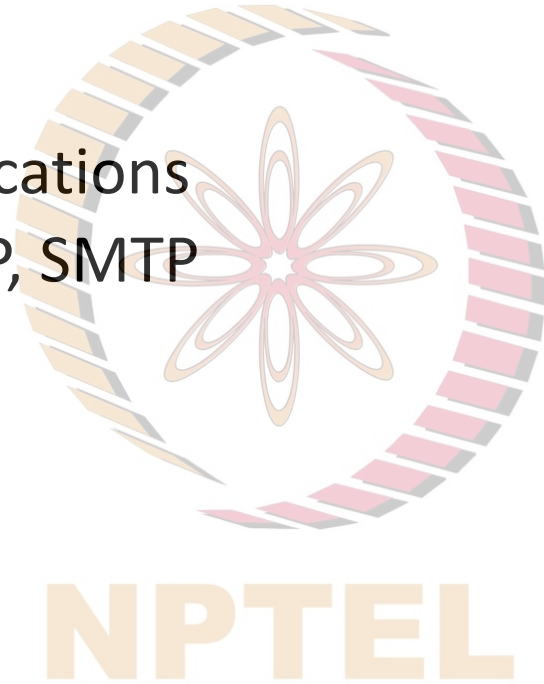
- ☐ Data formatting & translation
- ☐ Encryption / Decryption
- ☐ Compression



OSI Layers –Application Layer

☐ Application Layer

- ☐ Interface for user applications
- ☐ Protocols like HTTP, FTP, SMTP



Layered TCP/IP Stack

- ❑ *application*: supporting network applications
 - ❑ HTTP, IMAP, SMTP, DNS
- ❑ *transport*: process-process data transfer
 - ❑ TCP, UDP
- ❑ *network*: routing of datagrams from source to destination
 - ❑ IP, routing protocols
- ❑ *link*: data transfer between neighboring network elements
 - ❑ Ethernet, 802.11 (WiFi), PPP
- ❑ *physical*: bits “on the wire”

